

DIVYARTH JAIN

Electronics & PCB Design · Embedded Systems · Robotics

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EDUCATION

VIT Bhopal University, B.Tech Electronics & Communication Engineering (AI & Cybernetics) Sept 2023 – present
Model Higher Secondary School, Senior Secondary (PCM) 2023

PUBLICATION

A. Chouksey, D. Joshi, **D. Jain**, A. Beohar, *et al.*, “Stability-Aware NC-CMOS Logic Design Through Ferroelectric Capacitance Matching: A Landau–Khalatnikov-Based Parametric Study.” *Accepted, to appear* (Paper ID 2381).
• One sizing rule for hysteresis-free negative-capacitance logic, $r = C_{\text{MOS,max}}/|C_{\text{FE}}| < 1$, verified in ngspice (PTM 45 nm BSIM4, Landau–Khalatnikov ferroelectric): it matched 267/270 PVT-corner and 1952/2000 Monte-Carlo runs.

PCB & ELECTRONICS

Six-Servo Driver and Sensor-Feedback Board Sept 2026
KiCad 10, PCA9685, MCP6004, MCP3008, Raspberry Pi

- Schematic for a robot-arm board: 6-channel PWM drive over I²C, plus 12 feedback channels (joint position and motor current) over SPI.
- Current sense: 0.15 Ω shunt into a non-inverting stage, $G = 1 + 19.1\text{ k}/1\text{ k} = 20.1$, so 1 A maps to 3.02 V inside the 3.3 V ADC range; 1 k/100 nF filter at $f_c \approx 1.6\text{ kHz}$.
- Power: 5 A polyfuse, IRF4905 reverse-polarity FET, SMBJ5.0A TVS, 1000 μF bulk (1 A per servo). Rev 3; layout next.

Prometheus S3 board (EasyEDA, 4-layer, JLCPCB rules): ESP32-S3 with two USB-C ports (native USB + CP2102N UART), TVS ESD protection, auto-programmer flashing. **Also:** ESP32 ToF distance meter (VL53L0X, OLED, RGB status) · AC energy meter with surge auto-cutoff · Verilog motor controller on FPGA vs. MCU-interrupt control.

ROBOTICS

Capstone (DSN4091): Physics-Aware Robot Brain 2026 – present
ST3215-HS bus servos, Raspberry Pi 4B, Pi Cam 2, PyTorch, Isaac Sim, MuJoCo

- An on-board world model predicts the next state and retrains itself only when its prediction error (“surprise”) spikes.
- Simulation, 10 seeds: after a sudden change, surprise drops to 42 vs. 114 for a frozen model, using about 4× less training compute than always-on learning (376 vs. 1570 gradient steps).

ArcheoHex Hexapod Feb – Oct 2024
18 servos, Raspberry Pi, Python, inverse kinematics, MobileNet

- Built an 18-servo hexapod for tight, uneven cavities; wrote the inverse-kinematics engine from scratch for gait adaptation on rough ground.
- On-board MobileNet object detection at 88.7% accuracy in low light.

Hexapod v2 Dec 2024 – Aug 2025 (approximate)
RP2040 Zero, 2× PCA9685, ESP32, 3 UBECs, 3D-printed frame

- Ported open-source hexapod firmware to my own controller stack and added smoother gaits; ESP32 runs the HUD and an Xbox controller link, three UBECs keep the servo rail stable.

Eco-SLAM Rover
ROS2 Humble, Nav2, ESP32, Raspberry Pi 4, 8× VL53L0X

- 360° ToF ring instead of LiDAR (about 80% cheaper); custom ROS2 driver feeds the 8 streams to Nav2.

EXPERIENCE & LEADERSHIP

QSC Solutions, AI Product Intern – Multi-agent AI May – Jul 2026
Autonomous multimodal multi-agent framework: fine-tuned language models (incl. PEFT) for a multilingual voice avatar that answers in the caller’s dialect.

QSC Cloud (QSC Solutions), AI Product Intern – Video Personalization May – Jun 2025
AI Video Avatar product: integrated AI models, improved personalization logic, helped build the ML training module.

IEEE IECS 2026, VIT Bhopal, Vice Chair, Student Volunteer Committee Jul 2026
Led student volunteers for VIT Bhopal’s first IEEE conference; designed its posters and banners.

A.I.E.M. Club, VIT Bhopal, President Jul 2024 – present
Workshops on PCB design, embedded AI and robotics for 100+ students; teaches juniors ESP32-S3/C3 in MicroPython.

SKILLS

PCB & EDA	KiCad 10 (schematic, layout, libraries), LTspice, ngspice (BSIM4), Cadence/Synopsys with SCL 180 nm PDK (lab setup)
Embedded	ESP32 (ESP-IDF, Arduino, MicroPython), STM32 HAL, FreeRTOS, Raspberry Pi; I ² C, SPI, I ² S, UART, CAN
HDL & Code	Verilog, VHDL, C, C++, Python, MATLAB, Bash
Robotics & ML	ROS2, Nav2, Isaac Sim, MuJoCo, PyTorch, inverse kinematics

ACHIEVEMENTS

Represented India at an international sailing championship; 3× national sailing, 1× national snooker · Caterpillar Autonomy Challenge finalist (IIT Madras, rover team) · MathWorks DSP certificates